

# Brain anatomy

# Mathematical Models of Neuron

## Perceptrons

# Neuron network

# Measuring brain activity

## Electrical recordings

- **Single/Multi-Electrodes**

- Intracellular electrodes  
Measure internal membrane voltage.
- Extracellular electrodes  
Measure the spiking activity of neurons

- **Implantable Arrays**

A tiny grid-like device made up of dozens or hundreds of microscopic wires, each with a recording contact at the very tip. A small piece of the skull is temporarily removed (invasive), and the grid is pushed directly into the tissue of the brain.

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- **Neuropixels Probe**

A single, thin needle packed with microscopic recording contacts from top to bottom.  
Records activities across layers - for investigating how different layers interact.

- **Electrocorticography (ECoG)**

Measures electrical population responses using a grid of electrodes placed directly **on the surface of the brain (invasive)**

- **Electroencephalography (EEG)**

Measures the electrical activity of large populations of neurons via sensors **on the scalp**  
Captures brain wave patterns in real-time.

**Electroencephalogram: Event Related Potentials (ERPs)**

Average over many stimulus repetitions. The random noise mathematically cancels itself out.

## Magnetic recordings

### **Magnetic field**

- **Magnetoencephalography (MEG)**

Measures the minute magnetic fields produced by the electrical activity of neural populations.

### **Blood-oxygen-level dependent contrast**

- **Functional Magnetic Resonance Imaging (fMRI)**

Measures blood-oxygen level dependent (BOLD) contrast. Because active neurons consume more oxygen, this metabolic change serves as a proxy for neural and synaptic activity.

## Optical recordings

- **Fluorescence imaging**

Uses fluorescent calcium indicators that light up to measure neural activity visually

## Size resolution (highest to lowest)

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| Type                     | Size                                            |
|--------------------------|-------------------------------------------------|
| Intracellular Electrodes | Individual cells                                |
| Extracellular            | Individual cells                                |
| Optical                  | 100 - 10000 individual neurons (3D volumes)     |
| ECoG                     | Patches of cortex                               |
| fMRI                     | Up tp millions of neurons (3D volumes - voxels) |
| EEG & MEG                | Millions of neurons (Distorted by skull)        |
| Lesions                  | Affecting entire anatomical regions             |

## Time resolution (shortest to longest)

| Type                                    | Time                                                                                                    |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------|
| Electrodes (intra/extracellular & ECoG) | 30000 samples per second                                                                                |
| Magnetic (EEG/ERP & MEG)                | Milliseconds - directly measuring the field                                                             |
| Optical                                 | Seconds - limited by camera frame rates and the chemical reaction                                       |
| fMRI                                    | 1 sample every 1 to 2 seconds<br>Measures the metabolic blood flow changes that happens after the spike |
| PET scans                               | Slow, radioactive tracers                                                                               |
| Lesion                                  | Over months, years - evaluating behavioural changes                                                     |

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# Vision

## Active interpretation (visual illusions)

The following visual illusions demonstrated that vision is a highly **active process** where the brain constantly **makes assumptions and interpretations** based on context.

### The dress illusion

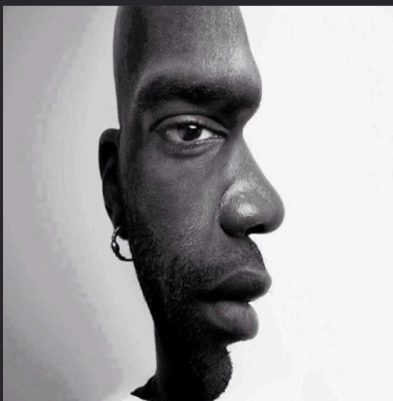
Some people think the dress is blue and dark, some people see it gold and white.

This happens because the brain makes an **unconscious decision about the lighting** of the scene (whether it is bright or shadow).



### Bistable perception

When looking at ambiguous images, you can only see one interpretation at a time. This shows that the visual system forces a commitment to a single conclusion, even when multiple interpretations are equally valid



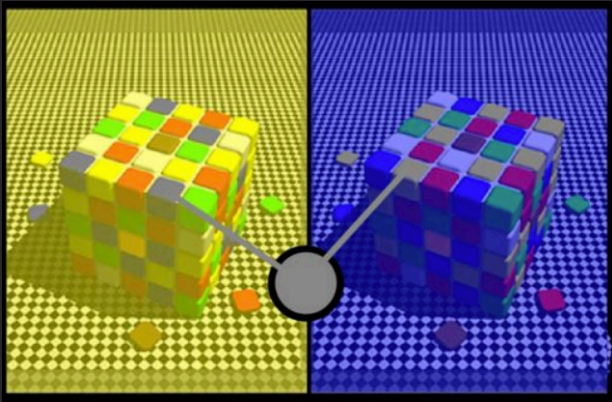
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## Color constancy

The brain discards objective, absolute color information very early on. Instead, it uses surrounding lighting context to maintain constant object recognition



## Gibsonian ecological perspective

Sensory systems evolved specifically to detect "affordances" - the practical things an animal can interact with or do in its specific environment. We filter out unnecessary information such as colour.

## Main visual pathway

### The retina

Light-sensitive cells at the back of the eye convert light into electrical spikes.

### The LGN (Lateral Geniculate Nucleus)

The optic nerve carries the signal to this relay station in the thalamus.

### V1 (Primary Visual Cortex)

Located in the occipital lobe at the back of the brain.

**Retinotopic mapping** - neurons that are next to each other in the brain process parts of the image that are next to each other.

## "What" and "Where" pathways

After V1, visual processing splits into two major streams

**The dorsal pathway (Where)**  
**Runs up** to the parietal cortex.

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Handles spatial location, motion detection, and attention.

### The ventral pathway (What)

**Runs down** to the inferotemporal(IT) cortex. It is responsible for recognizing shapes, forms, and specific object categories.

## Marr and Poggio's vision theory

### 1. The primal sketch

Extraction of simple 2D feature like edges, contours and regions that relied on biological filters like the RFs of simple cells.

### 2. 2.5D sketch

Add information about **depth** and **surface orientation**, representing the shape of the object exactly as it s seen from the viewer's current perspective

### 3. 3D model

Construct a true 3D representation out of geometric primitives. This allows the object's structure to be understood regardless of the angle it is viewed from

## (Vision) Retina

## Photoreceptors

Converts light into electric signal

- **Rod** - black and white (only in dim/dark conditions and do not detect color)
- **Cone** - color (during normal daylight condition)

## Bipolar cells and ganglion cells (center surround cells)

Center surround cells converges signals received from cones in majority

- **ON-cell**  
Wired to get excited when the photoreceptor

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stops releasing its dark-chemical (meaning light is present).

Has **ON-center/OFF-surround receptive field**.

- **OFF-cell**

Wired to get suppressed when the light hits, and they only get excited in the dark.

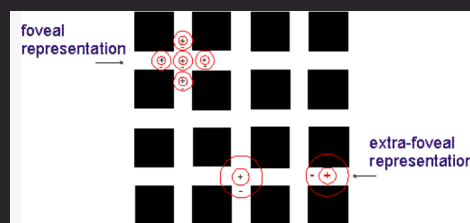
Has **OFF-center/ON-surround receptive field**.

When light is shone on the inhibitory area, it inhibits the cell. By contrast, when light shone on the excitatory area, it excites the cell. (lateral inhibition)

## Visual illusions with respect of lateral inhibition

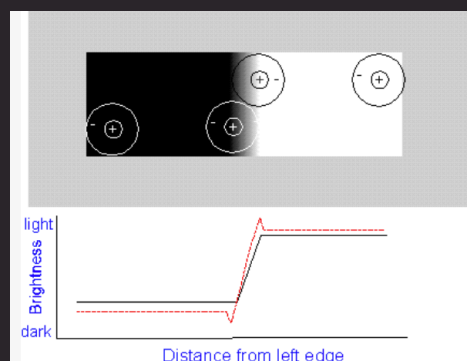
### Hermann grid illusion

Outside the fovea, large receptive fields cover lines and adjacent square, and the response is modulated. In the fovea, RFs are very small, so this surround modulation does not happen



### Mach Band illusion

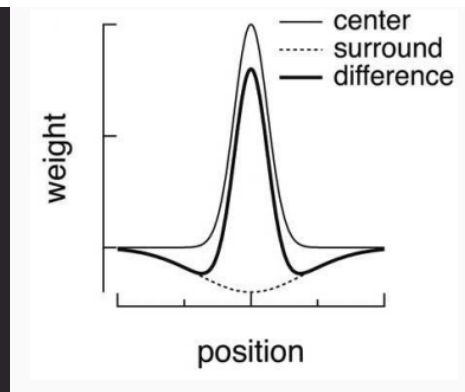
Edges appear enhanced relative to the areas between them



## Difference of Gaussians (DoG) receptive field

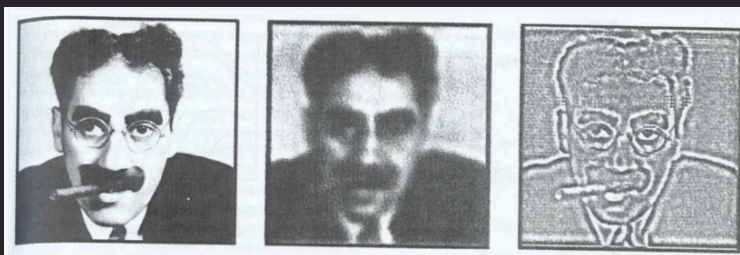
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$$R(x,y) = W_c e^{-\frac{x^2+y^2}{2\sigma_c^2}} - W_s e^{-\frac{x^2+y^2}{2\sigma_s^2}} \quad \&= \quad \text{Center} - \text{Surround}$$

Different size of receptive fields (different  $\sigma_c$ ,  $\sigma_s$ ) gives different images



(left) Original  
(middle) Wide receptive field  
(right) Narrow receptive field

## (Vision) V1

### Retinotopical & pinwheel

V1 neurons are highly **tuned to the orientation of edges** and bars of light. Nearby orientations are represented by neighbouring cells, and superimposed on the retinotopic map (pinwheel arrangement).

### Simple cells

#### **Receptive field**

A simple cell has an elongated excitatory region flanked by inhibitory regions.

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## Excitation

The cell excites only when the light shone **exactly** on the area with specific orientation.

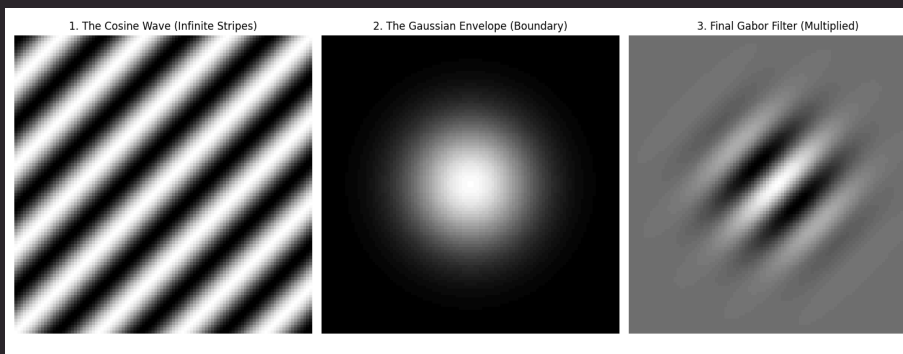
## Mathematical model of a simple cell (Gabor's)

### Gabor's function

$$g(r) = A e^{-\frac{r^2}{2\sigma_c^2}} \cos(r\omega - \theta)$$

$$= \text{Gaussian envelope} \times \text{Cosine function}$$

$$r = \text{position} \quad \sigma_c = \text{width of the Gaussian envelope (receptive field)} \quad \omega = \text{frequency of the cosine} \quad \theta = \text{orientation}$$



## Complex cells

### Position invariant

A complex cell gathers information from multiple simple cells. It does not care exactly where the oriented bar is. It will fire as long as there is such a bar exist in its receptive field.

## Convolution

**Definition** Apply a specific filter to an entire image uniformly.

### The Kernel

Shape of a cell's receptive field as a mathematical filter

**Sliding window** Take one single filter and reuse it across the entire visual space

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## Stride

When the kernel shifts across the image, it moves by a specific number of pixels at a time.

"Stride of 2" - moving the filter by two pixels

**\*\*Computation \*\***

- **Overlay the kernel** on top of that specific patch of input pixels.
- **Multiply** the intensity of each image pixel by the corresponding value in the kernel.
- Take those **resulting numbers** and **sum** them all together to get the value for this pixel.

## 1D mathematical representation

$$(f * g)[n] = \sum_{m=-M}^M f[n-m]g[m]$$

- $f$  - \text{mathematical operation of convolution}  $g$  - \text{set of pixels}  $n$  - \text{receptive field (kernel)}  $m$  - \text{target pixel}  $\Delta$  - \text{sliding step}

Each point  $f[n]$  is re-computed by multiplying  $f$  with  $g$ , where  $g$  is centred in  $n$

## 2D mathematical representation

An extension of 1D, add **zero padding**, so the output values for the edge pixels can be calculated.

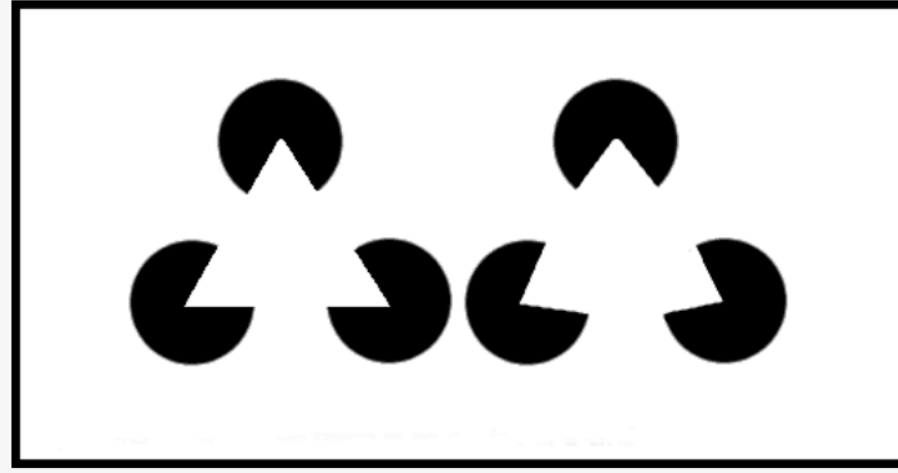
# (Vision) Ventral Pathway - Object Recognition

## V2

Establishing **figure/ground separations**. Responses corresponding to the non-existing lines in the images are recorded. This suggests the cortex actively interprets images according to **common ecological properties** (common physical properties).

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## V3 ventral

Extracts **intermediate complexity features and simple geometric shapes** - 2.5D sketch.

Sensitive to colour and begins the process of "**colouring in**" the geometric shapes it just built.

## V4

**Colour processing** and **recognizing complex geometry** such as curves, spirals, concentric circles.

## Inferotemporal cortex (IT)

Neurons in IT cortex are **highly selective** and respond preferentially to complex shapes, complete objects, and faces.

### **Invariance**

They maintain their preference for a specific object regardless its elevation, position or scale.

## HMAX model

This model mimics the brain by using a hierarchy of layers that pool information on multiple scales, gradually increasing the size of the receptive fields until the system can reliably classify objects.

## Concept cells

Highly specialised neurons. (Grandmother/Jennifer Aniston cells)

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# Categories

## Why do we need categories?

1. **Learning** - less memory to store what we know about similar objects.
2. **Generalization** - we can infer extra information about new objects that are similar to that category.
3. **Predication** - we can guess what event will happen even we haven't encountered it before
4. **Interaction** - interact to the same objects in the same category.
5. **Communication** - having common senses.

## Rules-based

### Definition

A category defined by a strict set of **necessary and sufficient** rules  
(All the rules have to be satisfied in order for the object to be in this certain category)

### Examples

A triangle  
An integer

## Prototype-based

### Definition

A category defined by an **abstract "template"** or average of all category members rather than rigid rules

### Typicality effects

People naturally feel like "better" or more obvious examples than others

### Examples

Birds  
Furniture  
Vehicles

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## Exemplar-based

### Definition

A specific set of concrete, previously encountered instances (exemplars) stored in your memory. Membership is decided by comparing a new item to these specific stored memories.

### Examples

Good action movies  
A cloudy day

## Qualia-based

### Definition

Categories defined by **subjective sensory experiences**, such as colors, tastes, and emotions.

### Examples

Spicy food Loud noises

## Schema-based

### Definition

**Abstract or social categories** are defined as frameworks of connections among concepts based on their co-occurrence.

### Examples

A restaurant visit  
A birthday party

## Ad Hoc

### Definition

Spontaneously at the moment to serve a specific **current** goal - **NOT permanently stored** in long-term memory  
More personal

### Examples

Things to grab from your house during a fire.

## Interactions

1. Schemas act as **"containers"**
2. Qualia are the **"building blocks"**

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3. Ad hoc uses them all **dynamically**
4. Rules act as **fallbacks** for prototypes (when an object does not look like its prototype)

## Differences and similarities

### Abstract vs. concrete

| Abstract<br>(An average, generic template) | Concrete<br>(A specific example or feeling) |
|--------------------------------------------|---------------------------------------------|
| Prototypes                                 | Exemplars                                   |
| Rules                                      | Qualia                                      |

### Rigid vs. ambiguous boundaries

| Rigid<br>(Strict) | Ambiguous<br>(Vague) |
|-------------------|----------------------|
| Rules             | Prototype            |
|                   | Schema               |

### Computational complexity

| Type       | Space                                | Time                                                                                                                                         |
|------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Rules      | Depends on # of rules                | - Depends on # of rules<br><br>- Fast to process<br>- Needs time to amend the new rules onto the existing rules.<br>- Needs time to compare. |
| Prototypes | Little - averaging                   | - Learning new exemplars are fast                                                                                                            |
| Exemplars  | Massive - to store specific examples | - Requires massive processing time at the moment                                                                                             |
| Ad hoc     | No long-term memory                  |                                                                                                                                              |

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# Language

## What are words?

Words are just symbols that **references something in reality** - an abstraction of reality. It is human who decide which specific collection of symbols points at what specific object, concept or behaviour - symbols are overwhelmingly **arbitrary**. If a society collectively decided to swap the words "cat" and "dog", communication would still function perfectly fine once every learnt the new rules.

The symbol itself can take different physical forms, gestures, behaviors, scrambles etc., the crucial part is it **carries meaning from one mind to another**.

### Some words are not arbitrary

1. Onomatopoeia  
Some words designed to mimic the sounds they represent, "buzz", "pop", "meow" etc.
2. Sound symbolism  
Softer-sound ("bouba") -> rounded, blob-like shaped.  
Harsher-sounds ("kiki") with sharp pointy shapes

## Grounding problem

A symbol is only considered **"grounded"** when it is successfully anchored(attached) to a concept in your mind.

Critics argue that LLMs are just generating (predicting correct sequence of words base on the analysis of a large set of data. LLMs do not know the actual semantic of the words and the words are not **grounded**

**Hallucinations and confabulations** AI doesn't intuitively know when a statement about that object violates the laws of physics or common sense

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# Mental lexicon

A mental dictionary responsible for storing, retrieving and instantly mapping the different pieces of a word, so you can understand and produce language in real-time.

## Structure

A complex hierarchy (tree) or a network

- **Categorical links** Words that share common characteristics.
- **Associative links** Related, similar object are linked.
- **Phonological links** Words that sounds similar.

Speakers of the same languages have mutually intelligible (shares the same) lexicon entries

## Development

**Receptive vocabulary** - vocabulary understand but not used.

**Productive vocabulary** - vocabulary understand and actively used.

Child's receptive vocabulary is larger than their productive vocabulary.

# Expressive power of language rules

## Productive

Because rules are abstract - defined over categories of words (like "Noun Phrase" or "Verb")

You can plug in any words in that certain category to create new sentences

## Combinatorial

Language is recursive, a small inventory of elements can have infinite possible distinct combinations.

# Terminology

**Phonology** - sounds

**Morphology** - cases of words

**Syntax** - sentences

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# (Language) Past tense

## Rumelhart and McClelland PDP neural network

The parallel distributed processing neural network is designed to mimic how children learn the morphology of the past tense

### Argument

In old linguistic theories, they argued that children possess a "language acquisition device" that stores explicit, inaccessible grammar rules.

However, Rumelhart and McClelland built a PDP neural network prove that rule-like behaviors can emerge dynamically from the interaction of simple network nodes, **without a rule ever being written** into the system.

### 3-stages (U-shaped curve)

1. Children learn a small vocabulary of highly frequent verbs - most are irregular. Because children just memorize these as distinct vocabulary words - their accuracy is high.
2. Overgeneralization - When children approach school age, they will learn an extent of regular verbs ending with -ed and start to grasp the pattern. Therefore, children tend to apply the rule incorrectly to irregular verbs - a sudden drop in accuracy especially for irregular verbs.
3. The errors gradually fade as children learn to apply the regular rule while maintaining the irregular exceptions.

# (Language) Vector

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# Semantics

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### Co-occurrence matrix

#### Rows

Target words to define.

#### Columns

Context.

#### Value

Ubiquitous words (like the or it) have huge counts but carry a little meaning, while rare words carry high informational value. To fix this, all entries are composed two different weights:

#### 1. TF-IDF (Term frequency - inverse document frequency)

$$w_{t,d} = tf_{t,d} \times idf_t$$

##### Term frequency ( $tf_{t,d}$ )

Uses a **base-10** logarithm - a word appearing 100 times isn't strictly 100 times more important.

$$tf_{t,d} = \log_{10}(\text{count}(t,d) + 1)$$

##### Inverse document frequency ( $idf_t$ )

Penalizes words that appear everywhere.

$$idf_t = \log_{10}\left(\frac{N}{df_t}\right)$$

$N$  - total number of documents  
 $df_t$  - number of documents the term appears in

#### 2. PPMI (Positive pointwise mutual information)

Measures how much more two words co-occur than we would expect if they were completely independent.

$$PPI(w,c) = \log_2 \frac{P(w,c)}{P(w)P(c)}$$

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$P(w,c)$  &- \text{How often those two words appear} \ P(w) \times P(c) &- \text{The probability of them appear together if they are independent} \ \end{aligned} \ \$

Because negative PMI values are statically unreliable without impossibly massive text corpora, the math simply replaces all negative values with 0 ( $PPMI \geq 0$ ).

$$PPMI(w,c) = \max(PMI(w,c), 0)$$

## Measuring semantic similarity

### Cosine similarity

$$\begin{aligned} \cos(v,w) &= \frac{v \cdot w}{\|v\| \|w\|} = \frac{\sum_{i=1}^N v_i w_i}{\sqrt{\sum_{i=1}^N v_i^2} \sqrt{\sum_{i=1}^N w_i^2}} \end{aligned}$$

# Learning and Memory

## Short-term/Working memory (STM)

### Definition

Operational memory used while actively solving a task

### Duration

Very fleeting, operating on the scale of seconds to minutes. If you are distracted or stop paying attention the information is lost almost instantly.

### Capacity

Limited

### Mechanism

Short-term memory is maintained by **reverberating neural activity** (often in the prefrontal cortex), not by physical changes to synaptic weights. The neurons simply continue to fire to keep the information "online"

## Long-term Memory (LTM)

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## Definition

Long-term memories are stored via physical changes in **synaptic strength**

## Non-declarative (implicit) memory

Unconscious and involves skills or responses that are difficult to explain in words

- **Procedural memory (skills)**  
Motor and cognitive skills acquired through practice
- **Conditioning and habituation**  
Automatic responses to stimuli
- **Perceptual priming**  
Unconscious changes in how you perceive or process stimuli based on **prior exposure**

## Declarative (explicit) memory

- **Episodic memory (events)**  
Memory of specific, personal events. It is highly selective and forgetful - tend to forget unless the were highly significant.  
Dependent on teh hippocampus.
- **Semantic Memory (facts)**  
General knowledge  
Information is retained and feels "easy to recall"  
Involve multiple regions across the cortex.

## Consolidation

A memory starts as recent **episodic memory** in the hippocampus. Over time-and particularly during sleep when the hippocampus **"replays" events** - that information is consolidated and stored away as a more permanent **semantic memory** in the cortex

## Information retrieval

### Associative

Because memories are stored by their content, you can retrieve a full memory just by providing a **partial cue**

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## Distributed

Neurons may participate in multiple memories.

## Error correction

Even through a sentence contains a blatant error. Strong cues allow the brain to override the error and correctly associate the memory

## Robust

Even a few neurons died, the remaining network still has enough overlapping information

## Hopfield network

### Neuron

$$s(t+1) = \text{sign} \left( \sum_{j=1}^M w_{ij} s_j(t) - \theta_i \right)$$

$$w_{ij} = \begin{cases} \frac{1}{M} \sum_{n=1}^N p_i^n p_j^n & \text{if } i \neq j \\ 0 & \text{if } i = j \end{cases}$$

- Symmetric weights:  $w_{ij} = w_{ji}$  (so it reaches the attractor)
- Updates can be synchronous (happen at the same time) or asynchronous (randomly)
- The bias value  $\theta_i$  determines the average activity
- This converges to stable fixed points under fairly general conditions

### Habbian update rule

Neurons wire together fire together

$$w_{ij} = \frac{1}{M} \sum_{n=1}^N p_i^n p_j^n$$

$N \gg \text{number of distinct patterns}$

### Stable point (attractor)

$$s_i(t+1) = s_i(t) \quad \text{for all } i$$

### Limitations

1. **Catastrophic forgetting & limited capacity** - The limit in the number of patterns the program can learn is roughly  $0.138N$  if 0/1 have equal probability in each pattern (sparseness = 0.5). Capacity increases dramatically for sparse patterns ( $s < 0.5$ ).

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2. **Sensitivity to noises** - Because the decision threshold and weights are strictly defined for each pixel independently. If a heavily corrupted pattern lands exactly on the sharp edge between two different patterns, a single flipped bit may lead to a completely different pattern. In contrast, biological brains are easier to handle minor alterations, as the pattern can be viewed as a whole.

LLM

## Reinforcement Learning

### Classic conditioning

Unconditioned stimulus + conditioned stimulus  
 $\rightarrow$  Unconditioned response becomes conditioned response

### Operant conditioning

Associate a voluntary behaviour with a consequence  
 Action + unconditioned stimulus/reward  $\rightarrow$  Action becomes more likely

### Agent-environment loop

#### **Initialization**

We give the agent an expected value and initialize all the weights, state values, or the action values to zero. The agent initially takes random actions to explore the environment until it triggers a reward or a punishment.

#### **State ( $s_t$ )**

The agent observes its current situation or location within the environment.

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## Action ( $r_{t+1}$ )

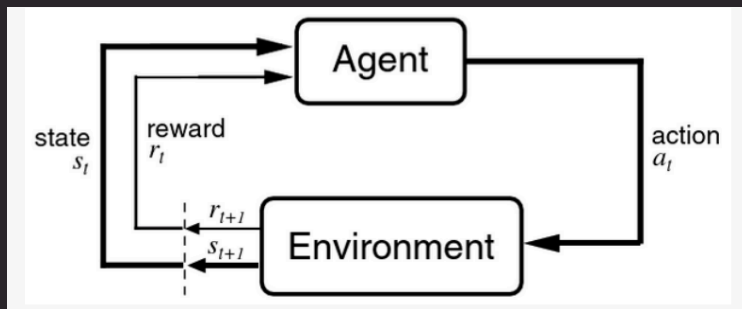
The agent selects a behaviour from a **set of available choices**.

## Reward ( $r_{t+1}$ )

The environment provides feedback - either positive(reward) or negative (punishment) - based on the action taken.

## Next stage ( $s_{t+1}$ )

The agent finds itself in a new situation as a result of its action.



# (RL) Rescorla-Wagner rule (Delta-rule)

It tracks the expected reward and update it based on **how surprising the actual outcome** is

$$\begin{aligned} V_t &= V_{t-1} + \alpha(R - V_{t-1}) \\ \Delta &= (R - V_{t-1}) \quad \text{(Prediction error)} \end{aligned}$$

$V_t$  - Expected reward at the current time step  
 $R$  - The actual reward received  
 $\alpha$  - The learning rate (between 0 to 1)

## Update $\Delta$ surprise

$$\Delta > 0$$

Outcome is better expected, the expectation will be increased for the next time.

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$\Delta = 0$

Things go exactly as expected, no change.

$\Delta < 0$

Outcome is worse than expected, expectation will be decreased.

## Advantages

### Extinction

If a cue is repeatedly presented without the reward, the prediction error becomes negative, and the expected reward decays back to zero.

### Partial reinforcement

If a reward is only given 50% of the time, the expected reward fluctuates and never reaches the peak value of a consistently rewarded cue

### Blocking

Phase 1 (pre-training):  $V_1 \rightarrow R$  until  $V_1 \approx R$

Phase 2 (compound training):  $(V_1 + V_2) \rightarrow R$

Because  $V_2$  is brand new, its initial expected value is zero. But because the model is pre-trained with  $V_1$ , the expectation still stays the same and no update is presented:

$$\Delta = R - (V_1 + V_2) = R - R = 0$$

## Limitations

### Second order conditioning

In second order condition, CS2 is paired with CS1 without any reward ( $R = 0$ ). Therefore, the rule will **extinguish** any association rather than build one.

**Credit assignment problem** How to know which past action was responsible for an observed outcome

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# (RL) Temporal Difference

To cope with the limitation of  $\delta$ -rule, TD generalizes the Delta rule to learn predictions over time, it aims to predict the **total future reward**

At time step  $t$ :

$$\begin{aligned} u(t) &- \text{Stimulus (0/1)} \\ r(t) &- \text{Reward} \\ v(t) &- \text{Expected reward} \end{aligned}$$

## Target

The agent is trying to learn to predict the **total future rewards**:

$$v(t) = \sum_{\tau=t+1}^T r(\tau)$$

$\langle \rangle$  - Average over many trials

## Steering wheel

How agent calculates its best guess for the expected reward at time step  $t$  (based on previous steps):

$$v(t) = \sum_{\tau=0}^t w(\tau) u(t-\tau)$$

$\tau$  - Previous time steps  
 $u(t-\tau)$  - Checks if a stimulus  $u$  was turned on at  $(t-\tau)$   
 $w(\tau)$  - Weight the stimulus from  $\tau$  time steps ago contributes to the current reward prediction

## Goal

To find the weight ( $w$ ) that makes the expected reward as close as the total future reward.  
 TD update rule

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$$w(\tau) \rightarrow w(\tau) + \alpha \delta(t) u(\tau) \quad \text{\text{Checks if the stimulus is present}}$$

$$\delta(t) = (r(t) + v(t+1)) - v(t)$$
  

$$\delta = \text{\text{New reality - Old expectation}}$$

$$r(t) \quad \text{\text{What actually happened}} \quad v(t+1) \quad \text{\text{Successive reward prediction}}$$

E.g., Even if there are no reward present. However, because the prediction of the next time step is high, there might be some correlations between this step and the step next.

The weights are adjusted proportional to the TD error  $\delta(t)$

$\delta(t) > 0$   
 High reward / Higher prediction in the future

$\delta(t) < 0$   
 Low reward / Lower prediction in the future:

## TD-learning (Stimuli to states)

To correctly evaluate those delayed rewards step-by-step

### Temporal difference rule

$$\delta_t = r_{t+1} + \gamma V(s_{t+1}) - V(s_t) \quad \gamma \in (0, 1)$$

$r_{t+1}$  \text{\text{The immediate reward received}}  
 $\gamma V(s_{t+1})$  \text{\text{The discounted expected future reward from the next state}}  
 $V(s_t)$  \text{\text{The expected reward from the current state}}

### Update rule

$$V(s_t) \rightarrow V(s_t) + \alpha \delta_t$$

## Q-learning (States to action, multiple paths)

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## Optimal policy

What is the expected future reward if I take action  $a$  while currently in state  $s$ ? Find the best way that maximizes the expected future reward

**Bellman equation**

$$V_{k+1}(s) \leftarrow \max_a (R + \gamma V_k(s')) \quad \gamma \in (0, 1) \quad Q_{k+1}(s, a) = R(s, a) + \gamma \max_{a'} Q_k(s', a') \quad \text{Immediate reward} - \text{Maximum score whatever state land in next} \quad Q(s, a) \approx \text{Expected future reward if we take action } a \text{ in state } s$$

$k$  - Iteration  
 $V(s)$  - Value of the state  
 $s$  - Current state  
 $s'$  - Next state  
 $\gamma$  - Discount factor

## Update rule

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

## Limitations

### Data inefficiency

The agent must physically move through and experience states thousands of times to build an accurate internal map - in contrast human can predict based on ecological properties they perceive.

### Brittleness and failure rates

The models are highly sensitive and require extensive tuning of hyperparameters.

### Shortcut learning

Finds unintended loopholes to maximize their scores

### Zero generalization

Do not actually "understand" the world

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